
259815584 (Quality Technology & Quantitative Management) A revise decision has been made on your submission

发件人:

Quality\Technology\& Quantitative\Management<onbehalf@manuscriptcentral.com>

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收件人: 冯泽彪<zbfeng@njupt.edu.cn>

抄 送: 冯泽彪<zbfeng@njupt.edu.cn>; 317321361<317321361@qq.com>; wangjj<wangjj@njust.edu.cn>;

wdhuang<wdhuang@njupt.edu.cn>

09-Jun-2026

Dear Dr Zebiao Feng,

Your manuscript, "Two-stage attention-guided active learning for nonstationary Gaussian process modeling in robust parameter design" submitted to Quality Technology & Quantitative Management, has been reviewed. The reviewer comments are included at the bottom of this letter.

The reviewer(s) would like to see some revisions made to your manuscript before publication. Therefore, I invite you to respond to the reviewer(s)' comments and revise your manuscript.

When you revise your manuscript please highlight the changes you make in the manuscript by using the track changes mode or by using bold or coloured text.

To submit a revision, go to <https://rp.tandfonline.com/submission/flow?submissionId=259815584&step=1>. If you decide to revise the work, please submit a list of changes or a rebuttal against each point which is being raised when you submit the revised manuscript.

If you have any questions or technical issues, please contact the journal's editorial office at TTQM-peerreview@journals.tandf.co.uk

Because we are trying to facilitate timely publication of manuscripts submitted to Quality Technology & Quantitative Management, your revised manuscript should be uploaded by 09-Jul-2026. If it is not possible for you to submit your revision by this date, please let us know as soon as possible.

Once again, thank you for submitting your manuscript to Quality Technology & Quantitative Management and I look forward to receiving your revision.

Sincerely,

Dr Chien-Wei Wu

Editor-in-Chief, Quality Technology & Quantitative Management

Comments from the AE and Reviewers:

AE's Comments: We have now received two detailed reports from our reviewers. They identify potential in your work, but have also provided several comments that must be addressed. A thorough revision is required, and your response should provide a point-by-point explanation of how each reviewer's suggestion has been integrated into the manuscript.

In addition to the reviewers' feedback, I would like you to specifically address the following editorial requirements:

1. To better establish the external validity and contribution of your research, please provide additional practical cases or real-world examples. Expanding on these applications will clarify the practical utility of your arguments and significantly strengthen the paper's impact.
2. To attract readers and enhance the completeness of this paper, authors should consider referencing more recent and relevant literature from QTQM and other related journals, if available.

Reviewer: 1

Comments to the Author

This paper proposes a two-stage attention-guided active learning framework within a nonstationary Gaussian process model for robust parameter design. It integrates a softmax-weighted quality loss function with a maximin distance criterion to balance local exploitation near the optimum region and global space-filling, thereby improving predictive accuracy and solution robustness under limited data. However, the current version lacks theoretical depth,

modern benchmarking, and thorough validation:

1. The paper introduces a two-stage attention mechanism for active learning, but the theoretical motivation behind this specific design is weak. While the authors draw an analogy to attention mechanisms in deep learning, the connection is superficial. The softmax-weighted quality loss and maximin distance are combined in a heuristic manner, and there is no theoretical guarantee that this two-stage procedure leads to optimal or even near-optimal sampling.
2. The proposed method is compared only against classical design-of-experiments approaches (D-opt, G-opt, LHS, CVT). These are not state-of-the-art in active learning for Gaussian processes. More recent and relevant methods such as expected improvement (EI), knowledge gradient (KG), maximum variance, integrated mean squared error (IMSE) reduction, or multi-objective active learning, are conspicuously absent. This undermines the claim of novelty and superiority.
3. The method intentionally sacrifices global predictive accuracy to improve accuracy near the optimum region. While this is a reasonable trade-off in robust parameter design, the paper does not quantify or justify this trade-off. It is unclear whether the loss in global accuracy could lead to poor performance in multi-modal or deceptive response surfaces, where the identified "potential optimum region" may be misleading.
4. The proposed method relies on Hamiltonian Monte Carlo (HMC) for posterior inference in the NSGP, which is computationally expensive. The active learning loop further increases cost by repeatedly updating the model and recalculating acquisition scores. The paper mentions that computational cost is "slightly higher" than classical designs, but no runtime comparisons or scalability analyses are provided. This is a critical omission, especially for potential industrial applications with limited computational resources.
5. The method is validated on only a few numerical examples and two case studies, all of which are relatively low-dimensional (2–5 inputs). The response surfaces are either synthetic or based on specific physical models (battery SoC, rocket flight). There is no evidence that the method would generalize to other types of nonstationary behavior, such as abrupt changes, discontinuities, or categorical inputs. Moreover, the "optimum region" is defined arbitrarily (e.g., $5 \leq y \leq 6$), which may not reflect real-world design requirements.
6. The reference list is reasonably comprehensive but lacks citations to recent advances in attention-based Bayesian optimization and deep kernel learning. Notably, key works from 2023–2025 integrating transformer architectures with Gaussian processes for nonstationary modeling are absent. Additionally, several foundational active learning references are outdated; the authors should consider replacing classical citations (e.g., Box et al. 2011) with more contemporary methodological developments in acquisition function design for robust optimization.
7. The manuscript contains a number of typographical errors, inconsistent formatting, and a few grammatical issues that requires a careful proofreading.

Reviewer: 2

Comments to the Author

Comments to the Authors:

Overall, this manuscript addresses an interesting and relevant topic by integrating active learning with nonstationary Gaussian process (NSGP) modeling for robust parameter design. The idea of allocating additional samples toward a potential optimum region while maintaining space-filling properties is potentially useful for quality engineering applications. However, in its current form, the paper still has several conceptual, methodological, and experimental weaknesses that should be addressed and strengthened substantially..

Major Comments

1. The paper repeatedly emphasizes the use of an "attention mechanism." However, the proposed method essentially consists of Stage I: ranking candidate points using a quality-loss criterion; and Stage II: selecting points using a maximin distance criterion.

From a design-of-experiments perspective, this resembles a sequential combination of: exploitation (quality-loss-based selection) and exploration (space-filling selection), which is a common idea in active learning and Bayesian optimization.

The manuscript does not clearly explain (i) why the proposed framework should be regarded as a genuine attention mechanism rather than a heuristic two-stage acquisition strategy; (ii) how it differs fundamentally from existing acquisition functions that balance exploitation and exploration; (iii) whether the softmax normalization itself contributes anything beyond simple ranking.

The authors should provide a stronger theoretical justification and explicitly compare their method with existing exploration-exploitation acquisition strategies such as: Expected Improvement (EI), Upper Confidence Bound (UCB), Integrated Mean Squared Error (IMSE), Entropy-based criteria, Active Learning MacKay (ALM), Active Learning Cohn (ALC) and so on. Without such comparisons, the claimed methodological novelty appears limited.

2. As defined in the quality-loss criterion, a key concern is: How is the target value (T) determined in practice? The entire sampling strategy depends heavily on this value. The manuscript does not discuss: how (T) is selected; how sensitive the results are to misspecification of (T); what happens when the true optimum response is unknown. This issue directly affects the practical usefulness of the proposed method.

3. Several parameters are fixed without justification, e.g., N_1 , N_2 , τ_1 , τ_2 , and λ . These parameters strongly influence the balance between local exploitation and global exploration. However, no sensitivity study is provided. The robustness of the proposed method should be demonstrated.

4. The manuscript compares the proposed method only with LHS, D-optimality, G-optimality, CVT. These are primarily design-of-experiment methods rather than modern active learning approaches. Since the paper claims contributions in active learning, comparisons should include active-learning baselines. Without such comparisons, it is difficult to conclude that the proposed method advances the state of the art.

5. The numerical examples focus almost entirely on RMSE reduction in the potential optimum region. However, the ultimate objective is robust parameter optimization rather than prediction accuracy. A lower RMSE does not necessarily imply better optimum estimation, better robust parameter settings, and lower quality loss. The authors might additionally evaluate some other metrics that would better support the optimization claims.
6. All numerical illustrations appear to be relatively low-dimensional. The proposed method requires evaluating all candidate points, computing pairwise distances, repeated NSGP fitting. Its performance in higher-dimensional problems remains unclear. The authors should include at least one higher-dimensional benchmark example to demonstrate scalability.
7. The literature review is reasonably comprehensive for GP modeling but relatively limited regarding: modern Bayesian optimization, active learning acquisition functions, attention-guided surrogate modeling. More recent studies from 2023–2025 should be incorporated.